Smart Parking Fee Management System

1. Problem Statement

Develop a Python program to calculate parking fees based on the type of vehicle and the duration of parking. The system should accept vehicle details, calculate the parking charge, and display the total amount payable.

2. Objectives

- To calculate parking fees based on vehicle type.
- To calculate parking duration.
- To apply different rates for different vehicles.
- To calculate the total parking fee.
- To generate a parking receipt.

3. Assumptions

The following parking rates are considered:

Vehicle Type	Parking Fee per Hour
Two Wheeler	₹10
Car	₹20
Bus	₹50
Truck	₹40

Additional assumptions:

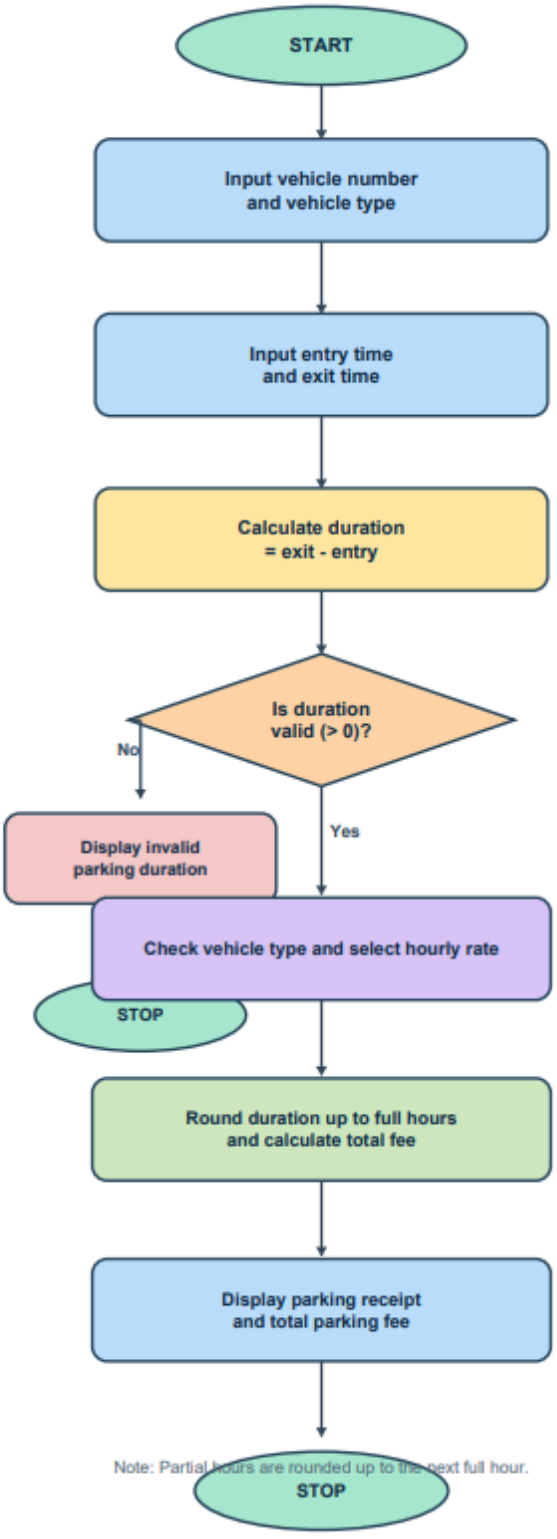
- Minimum parking duration is one hour.
- Any additional fraction of an hour is rounded up to the next full hour.
- Parking duration must be greater than zero.

- Vehicle type must be valid.
- These rates are assumed for academic purposes.

4. Algorithm

1. Start.
2. Input vehicle number.
3. Input vehicle type.
4. Input parking entry time and exit time in hours.
5. Calculate the parking duration.
6. Round up the duration to the next full hour.
7. Check the vehicle type.
8. Select the corresponding hourly parking rate.
9. Calculate the total parking fee.
10. Display vehicle details and parking fee.
11. Stop.

5. Flowchart



None

6. Pseudocode

```
None
START

INPUT vehicle_number
INPUT vehicle_type
INPUT entry_time
INPUT exit_time

duration = exit_time - entry_time

IF duration <= 0:
    DISPLAY "Invalid Parking Duration"

ELSE:
    hours = CEILING(duration)

    IF vehicle_type == "two wheeler":
        rate = 10
    ELSE IF vehicle_type == "car":
        rate = 20
    ELSE IF vehicle_type == "bus":
        rate = 50
    ELSE IF vehicle_type == "truck":
        rate = 40
    ELSE:
        DISPLAY "Invalid Vehicle Type"

    fee = hours * rate

    DISPLAY vehicle_number
    DISPLAY hours
    DISPLAY fee

STOP
```

8. Sample Output

```
None
Enter Vehicle Number: AP39AB1234
Enter Vehicle Type: car
Enter Entry Time: 10
Enter Exit Time: 13.5

----- PARKING RECEIPT -----
Vehicle Number: AP39AB1234
Vehicle Type: car
Parking Hours: 4
Hourly Rate: ₹ 20
Total Parking Fee: ₹ 80
```

9. Test Cases

Test Case	Vehicle Type	Duration	Expected Fee
1	Two Wheeler	2 hours	₹20
2	Car	3 hours	₹60
3	Bus	2 hours	₹100
4	Truck	4 hours	₹160
5	Car	1.5 hours	₹40
6	Car	Invalid duration	Error message

10. Conclusion

The Smart Parking Fee Management System calculates parking charges based on vehicle type and parking duration. It selects the appropriate hourly rate, rounds partial hours upward, and generates a parking receipt. This program demonstrates the use of dictionaries, conditional statements, mathematical functions, and user input in Python.